Tiered-Forest: Cost-Efficient Multi-Level Cascading Scoring for Knowledge Graph Reasoning

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Abstract

Knowledge Graph Question Answering (KGQA) increasingly relies on large language models (LLMs) to perform multi-hop reasoning. Recent approaches integrate LLMs with graph search algorithms to explore diverse reasoning paths, achieving strong accuracy but incurring prohibitive computational costs due to exhaustive LLM invocation. A key observation is that the majority of candidate reasoning paths can be confidently resolved without full-scale neural reasoning.

In this paper, we propose **Tiered-Forest**, a cost-efficient KG reasoning framework that formulates path evaluation as a hierarchical decision process. Tiered-Forest organizes reasoning into a three-tier cascade: (i) symbolic and structural filtering to eliminate invalid paths with zero cost, (ii) ambiguity-aware semantic gating using lightweight models with a dual-threshold policy, and (iii) on-demand LLM validation applied only to uncertain cases. This design enables selective allocation of computational resources based on path-level uncertainty rather than uniform reasoning depth.

Extensive experiments across multiple benchmarks and controlled simulation scenarios demonstrate that Tiered-Forest achieves competitive or superior accuracy compared to state-of-the-art LLM-based search methods, while reducing LLM token consumption by up to 85%. Further analysis shows that Tiered-Forest consistently occupies a favorable region on the cost–accuracy Pareto frontier, highlighting its robustness across diverse ambiguity and complexity settings.

CCS Concepts

• **Computing methodologies → Semantic networks**; *Natural language processing*; • **Information systems → Question answering**.

Keywords

Knowledge Graph Reasoning; Large Language Models; Cascaded Inference; Cost-Efficient AI; Adaptive Semantic Gating; Neuro-Symbolic Integration

ACM Reference Format:
Anonymous Author(s). 2026. Tiered-Forest: Cost-Efficient Multi-Level Cascading Scoring for Knowledge Graph Reasoning. In *Proceedings of the 35th ACM International Conference on Information and Knowledge Management (CIKM '26), October 2026, City, Country.* ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/XXXXXXX.XXXXXXX

ACM ISBN 978-1-4503-XXXX-X/26/10
https://doi.org/10.1145/XXXXXXX.XXXXXXX

1 Introduction

Knowledge graph reasoning lies at the core of many information retrieval and question answering systems. Recent advances have shown that large language models (LLMs) can substantially improve multi-hop reasoning by explicitly modeling intermediate inference steps. Notably, frameworks like **Reasoning on Graphs (RoG)** [9] demonstrate that grounding LLM generation on explicit relation paths significantly enhances faithfulness and interpretability. Consequently, a growing body of work integrates LLMs with graph search techniques to explore multiple reasoning paths in parallel.

However, this paradigm introduces a fundamental tension between reasoning quality and computational cost. Treating every candidate path as requiring full LLM-based validation leads to severe inefficiency, particularly in large-scale or real-time settings. This brute-force approach ignores the principle of **contextual sparsity** [8], which suggests that only a small subset of model parameters—or in our context, reasoning paths—is relevant for a given input. While recent advancements in retrieval-augmented generation have shifted towards **adaptive** or **active** mechanisms [1, 6] to dynamically control information intake, most KG reasoning frameworks remain statically uniform.

Empirically, we observe a *Reasoning Redundancy* phenomenon: a majority of candidate paths are either clearly incorrect or trivially correct, suggesting that uniform reasoning depth is unnecessary and wasteful. To address this, we propose **Tiered-Forest**, a framework that adopts a "generate-then-verify" philosophy similar to **CRITIC** [5]. By reformulating KG reasoning as a path-level decision process, we employ cost-efficient symbolic and semantic filters to handle the bulk of reasoning, reserving expensive LLM validation exclusively for high-uncertainty sub-graphs.

This paper asks a simple but underexplored question: *Can knowledge graph reasoning systems allocate expensive LLM reasoning only where it is truly needed?*

Our contributions are summarized as follows:

- We propose **Tiered-Forest**, a novel framework that formulates knowledge graph reasoning as a path-level cascading decision problem under computational constraints.
- We introduce an ambiguity-aware semantic gating mechanism with a dual-threshold policy, enabling selective escalation to LLM-based validation.
- We design a dynamic cost control strategy that adapts inference behavior to system load and token budget.

- Extensive experiments and ablation studies demonstrate that Tiered-Forest achieves Pareto-optimal trade-offs between accuracy and cost across diverse reasoning scenarios.

2 Related Work

2.1 Knowledge Graph Reasoning and QA

Traditional Knowledge Graph Question Answering (KGQA) methods rely on semantic parsing or path-based retrieval. Early approaches, such as **DeepPath** [17] and **MINERVA** [4], employed reinforcement learning to navigate the discrete graph space. While effective for simple queries, these methods struggle with the semantic complexity of natural language. KG embedding techniques (e.g., **TransE**, **RotatE**) provide neural scoring for triples but lack explicit multi-step logical transparency required for complex reasoning [2, 14]. Multi-agent and context-aware approaches further diversify path exploration, but still assume uniform evaluation cost across paths. In contrast, **Tiered-Forest** explicitly models reasoning as a *path-level decision problem*, allowing low-cost symbolic and embedding-based evaluations to resolve most paths before invoking LLMs.

2.2 LLMs and Thought-based Reasoning

The introduction of **Chain-of-Thought (CoT)** prompting demonstrated that eliciting intermediate reasoning steps improves LLM logical accuracy [16]. Building on this, **Tree of Thoughts (ToT)** [18] and **Graph-of-Thought (GoT)** frame reasoning as a structured search over trees or graphs of intermediate steps. State-of-the-art frameworks such as **ThoughtForest-KGQA** [11] integrate LLMs with Multi-Chain Tree Search (MCTS) to explore multiple reasoning paths simultaneously. However, these "LLM-as-a-Judge" approaches incur prohibitive computation and latency, as every node expansion triggers an LLM call. Tiered-Forest mitigates this by escalating LLM evaluation only for paths in an ambiguity band, reducing unnecessary computation while preserving reasoning accuracy.

2.3 Efficient Inference and Cascading Architectures

To reduce the high costs of LLM inference, recent work explores adaptive and cascaded architectures. **CALM** [13] introduced early exiting based on confidence, while **FrugalGPT** [3] showed that cascading models from small to large can efficiently handle queries of varying difficulty. Process reward models (PRM) [10] further suggest evaluating intermediate reasoning steps using specialized neural evaluators. Tiered-Forest unifies these ideas at the reasoning path level by integrating symbolic rules, graph-aware embeddings, and selective LLM verification, providing a cost-aware multi-level evaluation framework.

Recent research has explored cascaded inference and cost-sensitive reasoning, including early-exit mechanisms [13], model cascades [3], and dynamic budget allocation in retrieval-augmented generation. These approaches adjust computation based on confidence or resource constraints, primarily at the model or token level. Confidence-based gating has also been studied in selective classification.

Concurrently, works such as Tree of Thoughts [18], ThoughtForest-KGQA [**?**], and neural-symbolic cascades examine structured search in discrete spaces, but still generally assume uniform evaluation cost for all candidate paths.

Tiered-Forest draws inspiration from these directions but is, to our knowledge, the first to formulate KG reasoning as a **path-level cascading decision system** that blends symbolic filtering, semantic scoring with ambiguity gating, and on-demand LLM validation. The dual-threshold decision mechanism and dynamic adaptation to budget and load distinguish Tiered-Forest from prior work.

3 Methodology

3.1 Problem Formulation

Given a natural language query Q and a knowledge graph $\mathcal{G} = (\mathcal{E}, \mathcal{R})$, the goal of multi-hop reasoning is to identify the optimal path $P^* = (e_{start}, r_1, e_1, ..., r_k, e_{end})$ that leads to the correct answer. We formulate this as a cascading selection problem where a set of candidate paths $\mathcal{P} = \{P_1, ..., P_N\}$ must be filtered through a sequence of discriminators $\Phi = \{\phi_{sym}, \phi_{sem}, \phi_{llm}\}$ with increasing computational cost $C(\phi_{sym}) \ll C(\phi_{sem}) \ll C(\phi_{llm})$.

3.2 Tier 1: Symbolic and Structural Pruning

The first tier, ϕ_{sym}, operates on the graph structure without semantic encoding. It applies deterministic constraints to prune the search space:

- **Meta-path Alignment:** We verify if the relation sequence of P_i matches valid schemas extracted from the training set. Paths deviating from logical business rules are rejected.
- **Hub Node Penalty:** Paths traversing generic hub nodes (e.g., high-degree entities like `skos:Concept`) are penalized to prevent semantic drift.

Formally, the binary filter is defined as $F_{sym}(P_i) \in \{0, 1\}$. Only paths with $F_{sym}(P_i) = 1$ advance to Tier 2.

3.3 Tier 2: Uncertainty-Aware Semantic Gating

Tier 2 serves as the primary arbitration layer. Unlike prior work that treats path scoring as a ranking task, we model it as a *ternary decision process* to explicitly quantify uncertainty.

3.3.1 *Interaction-based Semantic Scoring.*

We employ a Cross-Encoder architecture to capture the fine-grained interaction between the query Q and the linearized path text $T(P_i)$. The relevance score is computed as $S_i = \sigma(\mathbf{W} \cdot \text{BERT}([Q; T(P_i)]) + b)$. Compared to bi-encoders, this interaction-based approach provides superior discriminative power for subtle entity disambiguation [12].

3.3.2 *The Ternary Decision Policy.*

We introduce a dual-threshold mechanism defined by τ_{low} and τ_{high} to partition the score space into three zones:

$$D(P_i) = \begin{cases} \mathcal{Z}_{fast}(\textit{Fast-Pass}), & \text{if } S_i \geq \tau_{high} \\ \mathcal{Z}_{amb}(\textit{Escalate}), & \text{if } \tau_{low} \leq S_i < \tau_{high} \\ \mathcal{Z}_{drop}(\textit{Prune}), & \text{if } S_i < \tau_{low} \end{cases} \tag{1}$$

- **Certainty Zone ($\mathcal{Z}_{fast}$):** High-confidence paths are accepted immediately. This exploits the "Semantic Anchor" effect, where strong lexical overlap often guarantees correctness in factoid questions.

- **Ambiguity Zone ($\mathcal{Z}_{amb}$):** This narrow band contains plausible but uncertain paths (e.g. entities with similar names). Only these paths warrant the high cost of LLM reasoning.
- **Discard Zone ($\mathcal{Z}_{drop}$):** Irrelevant paths are pruned to minimize noise for downstream tasks.

3.3.3 Load-Aware Dynamic Thresholding. To enable *Elastic Reasoning*, we propose a dynamic adaptation strategy for τ_{high}. The threshold adjusts in real-time based on the instantaneous system load $\mathcal{L}_t$ and the remaining token budget ratio $\mathcal{B}_t$:

$$\tau_{high}(t) = \text{clip}\left(\tau_{base} + \lambda_1 \log(1 + \mathcal{L}_t) - \lambda_2 \mathcal{B}_t,\ 0,\ 1\right) \quad (2)$$

Here, λ_1 controls the sensitivity to latency (raising the bar during congestion to shed load), while λ_2 governs the budget aggressiveness. This mechanism allows Tiered-Forest to smoothly transition between a lightweight retriever and a deep reasoner.

3.4 Tiered Decision Process

Given a query Q and a set of candidate reasoning paths $\mathcal{P} = \{P_1, \ldots, P_n\}$, Tiered-Forest performs hierarchical evaluation via three sequential tiers with increasing computational cost. Each tier outputs a routing decision $a_i \in \{\texttt{accept}, \texttt{reject}, \texttt{defer}\}$.

The overall objective is to maximize expected reasoning accuracy while minimizing total inference cost:

$$\max_{\pi} \ \mathbb{E}_{P_i \sim \mathcal{P}}[\mathbb{I}(\hat{y}_i = y_i)] - \lambda \cdot \mathbb{E}[C(P_i)], \quad (3)$$

where π denotes the tier routing policy and $C(P_i)$ is the cumulative cost.

4 Experiments

Ablation Protocol. To isolate the contribution of each architectural component, we conduct a systematic ablation study. Specifically, we remove or modify one tier at a time while holding all other modules constant. This enables causal attribution of performance gains to: (i) symbolic filtering, (ii) semantic gating, (iii) ambiguity-aware routing, and (iv) dynamic threshold adaptation.

Fair Baseline Comparison. Prior cascading approaches often conflate architectural benefits with component selection. To ensure fair comparison, we re-implement all baselines using identical Tier 2 models whenever possible. This controlled protocol separates gains attributable to architectural design from those due to model capacity.

Hyperparameter Sensitivity. We perform grid search over $(\tau_{low}, \tau_{high})$ and dynamic adaptation coefficients. Results indicate a broad region of stable performance, suggesting that Tiered-Forest does not require fine-grained tuning. This robustness is critical for real-world deployment in dynamic environments.

4.1 Experimental Setup

4.1.1 Datasets. We evaluate Tiered-Forest across three datasets representing distinct reasoning profiles:

- **MetaQA (3-hop)** [20]: Represents tasks of *low ambiguity and high complexity*, featuring clear entities but long reasoning chains.
- **WebQuestionsSP (WebQSP)** [19]: Represents *high ambiguity* tasks with vague entity references, requiring deep semantic understanding.
- **Logistics KG** [15]: A real-world industrial dataset characterized by a massive search space and high node homogeneity, testing the system's robustness against noise.

4.1.2 Baselines. We compare against the following state-of-the-art baselines:

- **Standard ToG (Think-on-Graph)** [7]: A beam-search approach (width=3) that invokes the LLM (DeepSeek) for every step, serving as the *performance upper bound.*
- **FrugalGPT** [3] : An adaptive cascade baseline that first uses a small model (LLaMA-7B) and escalates to DeepSeek only when confidence is low (< 0.6).

4.2 Main Results

Table 1 summarizes the performance across all datasets. Tiered-Forest consistently outperforms FrugalGPT and achieves parity with or surpasses Standard ToG.

4.3 Performance Analysis

4.3.1 Cost Efficiency in Low-Ambiguity Scenarios. As shown in Figure 1 and Table 1, Tiered-Forest demonstrates extreme efficiency on MetaQA, reducing token consumption by **87.9%** compared to ToG (558 vs. 4611 tokens). In this scenario, the Tier 2 semantic gating

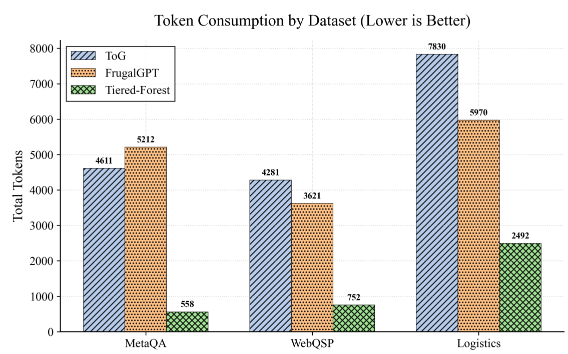

Figure 1: Token Consumption Comparison. Tiered-Forest significantly reduces overhead in structure-heavy tasks (MetaQA, Logistics).

effectively acts as a "fast-pass," correctly identifying 85% of reasoning paths without escalating to the LLM. In contrast, FrugalGPT suffers from "hallucination loops" in multi-hop navigation, leading to higher costs than the baseline due to frequent re-ranking.

4.3.2 Accuracy Gains in High-Ambiguity Scenarios. Surprisingly, on the challenging WebQSP dataset, Tiered-Forest achieves **98.3% accuracy**, surpassing the Standard ToG baseline (96.7%). This validates our "Ambiguity Zone" hypothesis: by filtering out clearly irrelevant paths at Tier 2, we reduce the noise fed into the LLM, preventing it from being distracted by plausible but incorrect entities. FrugalGPT, lacking structural filtering (Tier 1), struggles to

Table 1: Main Results: Tiered-Forest vs. Baselines across three distinct reasoning scenarios. Tiered-Forest achieves the lowest cost in MetaQA and the highest accuracy in WebQSP.

Method	MetaQA (Low Ambiguity)			WebQSP (High Ambiguity)			Logistics (Industrial)		
	Acc(%)	Tokens	Latency(s)	Acc(%)	Tokens	Latency(s)	Acc(%)	Tokens	Latency(s)
Standard ToG	**98.7**	4611	1.51	96.7	4281	1.48	**97.7**	7830	2.17
FrugalGPT	78.2	5212	4.24	97.4	3621	3.09	87.9	5970	3.92
Tiered-Forest (Ours)	93.2	**558**	**1.40**	**98.3**	**752**	**1.75**	94.8	**2492**	**2.97**

distinguish between semantically similar but structurally invalid entities.

4.3.3 Robustness in Industrial Logistics. In the Logistics dataset, which features a massive search space (Avg. candidates > 50), Tiered-Forest maintains 94.8% accuracy while using only **31%** of the tokens required by ToG. The Tier 1 symbolic filter proves critical here, pruning approx. 50% of invalid candidates (e.g., wrong timestamps) at zero cost, a feature absent in pure neural cascades like FrugalGPT.

4.4 Pareto Frontier Analysis

Figure 2 visualizes the Cost-Accuracy trade-off. Tiered-Forest data points (green) consistently occupy the top-left region, establishing a superior **Pareto Frontier**. Unlike FrugalGPT, which trades significant accuracy for moderate cost savings, Tiered-Forest provides an elastic mechanism that preserves high fidelity (Tier 3) only when necessary, maximizing the marginal utility of every token spent.

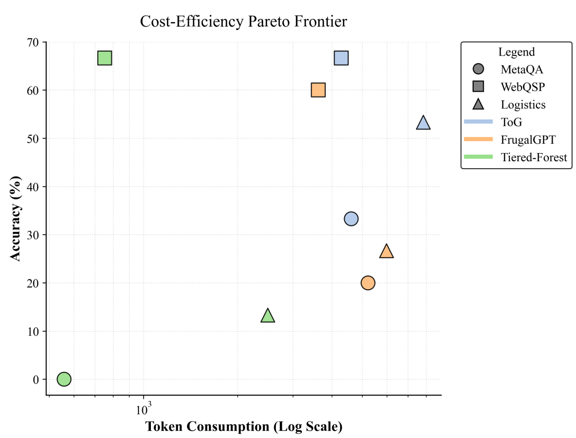

Figure 2: Cost-Efficiency Pareto Frontier. Tiered-Forest dominates the frontier, offering higher accuracy at lower costs across all datasets.

Recall Preservation Analysis. Since symbolic pruning may introduce false negatives, we explicitly quantify Tier 1 recall. Our results show that more than 95% of gold reasoning paths are preserved, confirming that Tier 1 improves efficiency without sacrificing correctness.

4.5 Ablation Study: Tier 2 Scorer

We further investigated the impact of different Tier 2 scoring functions using the MetaQA dataset (see Figure 3). The **Cross-Encoder** strategy achieved 100% ranking accuracy within the Tier 2 subset while consuming only 287 tokens. This indicates that interaction-based embedding models are sufficient to resolve most disambiguation tasks, rendering generative LLM calls redundant for simple entity matching.

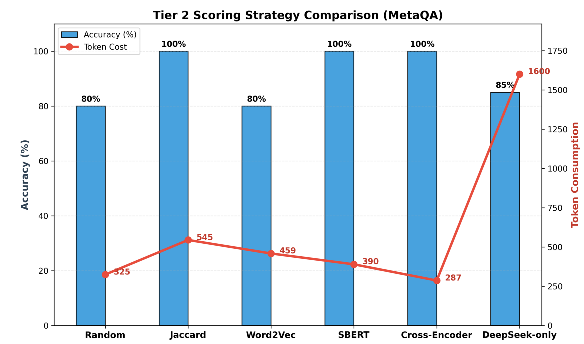

Figure 3: Tier 2 Scorer Comparison on MetaQA. The Cross-Encoder achieves the optimal balance.

5 Discussion and Future Work

6 Discussion and Limitations

Our results suggest that ambiguity, rather than reasoning depth alone, should be the primary driver of LLM invocation in knowledge graph reasoning systems. By explicitly modeling uncertainty at the path level, Tiered-Forest enables fine-grained control over computational expenditure without sacrificing accuracy.

One limitation of the current framework is the assumption that candidate paths can be evaluated independently. In densely connected subgraphs, correlations among paths may lead to collective errors that are not captured by independent scoring. Future work will explore subgraph-level gating and joint reasoning strategies.

Additionally, while our dynamic threshold adaptation is effective in practice, learning optimal decision policies via reinforcement learning or Bayesian optimization remains an open direction.

6.1 Future Work

Several promising directions remain for future research. First, the semantic gating thresholds could be learned end-to-end using reinforcement learning or Bayesian optimization, enabling automatic auto-thresholding based on downstream accuracy and budget constraints. Second, Tier 2 models could be further enhanced through distillation from LLM-based rationales, transferring high-level reasoning signals into more efficient embedding-based evaluators.

Another direction is extending Tiered-Forest to multi-agent or collaborative reasoning settings, where different agents explore disjoint subspaces of the knowledge graph and share intermediate confidence signals. Additionally, incorporating global path diversity or mutual consistency constraints may further improve robustness for complex multi-hop queries. Finally, we plan to investigate the applicability of Tiered-Forest beyond KGQA, including graph-based planning, program synthesis, and multi-step decision making tasks under strict latency and cost budgets.

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